

Evaluation of the high-performance TechnoTrak CR-39 for detection of 1-9 MeV fusion protons

N. Vanderloo,¹ A. T. Lanzrath,¹ E. Edwards,¹ T. E. Evans,¹ Z. Tessema,¹ M. Cufari,¹ S. G. Dannhoff,¹ B. C. Foo,¹ K. F. Bauer,¹ N. N. Keer,¹ B. I. Buschmann,¹ A. J. DeVault,¹ Y. Lawrence,¹ J. Vargas,¹ C. W. Wink,¹ N. R. Schwartz,¹ Shahina,¹ M. Gatu Johnson,¹ C. K. Li,¹ R. D. Petrasso,¹ and J. A. Frenje¹
Massachusetts Institute of Technology, Cambridge, MA 02139

(*Electronic mail: nielsv@mit.edu)

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In this paper, the MeV-proton response of a high-sensitivity, low-noise CR-39 nuclear track detector, TechnoTrak, produced by the Chiyoda Technol Corporation is evaluated. Plastic CR-39 nuclear track detectors are commonly used for precise time-integrated measurement of charged particles. Such measurements are crucial for diagnosing inertial confinement fusion (ICF) and high energy density (HED) experiments at the OMEGA Laser System, the Z Pulsed Power Facility, and the National Ignition Facility. The response of these TechnoTrak CR-39 detectors was tested with protons produced from DD and D³He fusion reactions at the MIT linear ion accelerator. The results indicate that TechnoTrak CR-39 detectors exhibit high detection efficiency over a wide proton energy range ($\sim 1 - 6$ MeV), remain sensitive to protons up to ~ 9 MeV, and have lower intrinsic noise than widely used TASTRAKTM CR-39 produced by Track Analysis Systems Ltd. This wide dynamic range and low noise floor may benefit future HED and ICF experiments requiring low background levels and broad proton energy sensitivity.

I. INTRODUCTION

Charged particle detection is critically important for diagnosing high energy density (HED) experiments, including inertial confinement fusion (ICF) and laboratory astrophysics experiments done on highly energetic laser and pulsed power facilities¹. Polyallyl diglycol carbonate, or Columbia Resin #39 (CR-39), is a plastic nuclear track detector^{2,3} widely fielded on HED experiments because of its high ion detection efficiency⁴ and relative insensitivity to x-rays^{5,6}, neutrons^{7,8}, and electromagnetic pulses common in such experiments. Diagnostics on the OMEGA Laser System⁹, the Z Pulsed Power Facility¹⁰, and the National Ignition Facility¹¹ that use CR-39 include magnetic recoil spectrometers^{12,13}, charged particle spectrometers^{4,14-19}, knock-on deuteron imaging¹⁶, and proton radiography.

Charged particles impacting CR-39 deposit energy, leaving a track of broken molecular bonds in the plastic. To reveal the damage, the CR-39 is placed in a bath of a heated alkaline solution (typically NaOH or KOH), which chemically etches away the track at a faster rate than the bulk material. Track sizes depend on the incident particle energy and etching conditions.

TASTRAKTM CR-39 produced by Track Analysis Systems Ltd.²⁰ is the only CR-39 currently used at the OMEGA Laser System, the Z Pulsed Power Facility, and the National Ignition Facility. Significant work has been done in the last thirty years to characterize TASTRAK CR-39 and qualify diagnostics that rely on it^{1,21}. In particular, the ion detection efficiency and diameter-to-energy relationship have been well studied using TASTRAK CR-39^{4,18,22-24}. Our understanding of the diameter-to-energy relationship is the basis for the analysis of the wedge-range-filter spectrometer^{17,18}. Work has also been done to study the robustness of TASTRAK CR-39 under typical operating conditions including vacuum exposure²⁵ and x-ray exposure^{5,6}. Small differences in CR-39 production method (e.g., dopants, chemical purity, and curing procedure) can cause changes in sensitive energy range, back-

ground noise levels, and best operating practices.

Intrinsic noise on CR-39 occurs when aberrations in the CR-39 due to, e.g., scratches or surface roughness, are falsely identified as a track. False tracks typically have high contrast, low diameter, and high eccentricity. Real tracks normal to the surface have near 0% eccentricity, but deviations occur due to particles impacting at an off-normal angle²⁶. However, most tracks in typical configurations have eccentricities between 0 and 15%. It is worth noting that if two or more tracks overlap with each other, they may be identified as one track with an abnormally high eccentricity. In all of our analysis in this paper, we only accept tracks with eccentricity under 15%, and do not expect significant track overlap at the attained fluence. It is not as easy to separate real and false tracks by diameter and contrast because real tracks exhibit a range of diameters and contrast values depending on etch length, particle energy, and particle type. The ratio of signal to noise in the regions in contrast-diameter space where real tracks are found is of significant interest.

In this paper, we study the performance of TechnoTrak2 (referred to here as TechnoTrak) CR-39 produced by Chiyoda Technol Corporation²⁷ under conditions relevant to HED experiments. Comparison is made to the commonly used TASTRAK CR-39. Limited study of TechnoTrak CR-39 has shown it to perform well for personal neutron dosimetry²⁸ and to have high sensitivity to low-linear energy transfer charged particles²⁹. Our results indicate that TechnoTrak CR-39 exhibits a wider dynamic range and lower intrinsic noise than TASTRAK CR-39. TechnoTrak CR-39 likely achieves low intrinsic noise levels by using a high purity (99.7%) monomer to create a smooth surface, and achieves wider dynamic range by the inclusion of an antioxidant ingredient IRGANOX1010 (0.01% wt)²⁹.

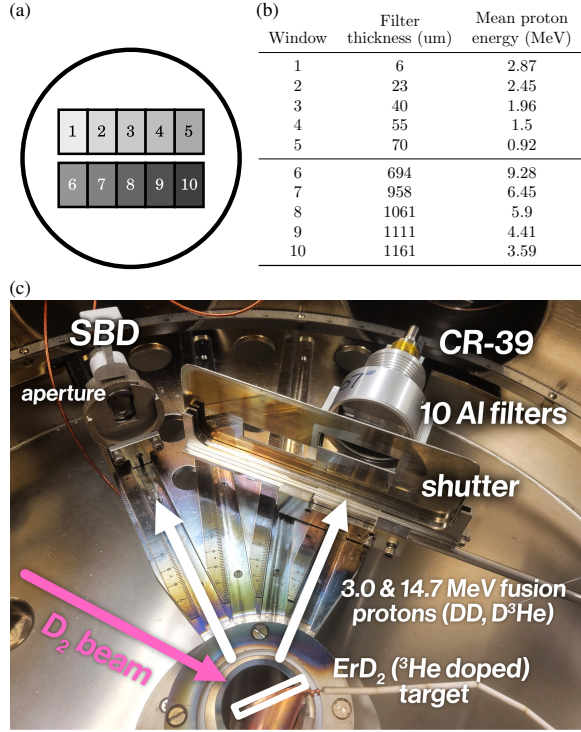


FIG. 1. (a) A schematic of the ten-step Al filter placed in front of the CR-39. (b) A table of the thicknesses of each window and the mean proton energy. (c) Photograph of an experimental setup inside the MIT linear ion accelerator target chamber. The CR-39 pack with the ten-step Al filter was positioned at 90° to the D₂ beam and 15 cm from the ³He-doped ErD₂ target, and the SBD was positioned at 141° and 18 cm from the target behind an aperture and a Ta filter.

II. EXPERIMENTS & RESULTS

A linear ion accelerator at MIT³⁰ was used to expose CR-39 to protons. Deuterons are accelerated by a 125 kV potential with the beam incident on a 30 μm-thick ³He-doped ErD₂ target. Deuterium-deuterium (DD) and deuterium-helium-3 (D³He) fusion reactions occur in the target, producing 3.0 and 14.7 MeV protons, respectively, in all directions. The rate of DD reactions in our experiments was ~20 times higher than that of D³He. Filter packs containing CR-39 are placed in the target chamber facing the target. A silicon surface barrier detector (SBD) is also placed in the chamber facing the target. The SBD actively resolves a proton energy spectrum and provides an independent measurement of the number of protons from both DD and D³He fusion events. Figure 1(c) shows the beam, target, CR-39, and SBD setup for one of the experiments described in this paper.

A. Intrinsic noise

A set of three experiments on the linear ion accelerator characterized the intrinsic noise of both TechnoTrak and TASTRAK CR-39 by exposing the CR-39 to monoenergetic protons with a background region. Circular pieces (5 cm diameter) of TechnoTrak and TASTRAK CR-39 were both placed 20 cm away from the target and at angles of 90° and 107° from the deuteron beam, respectively. An SBD was placed

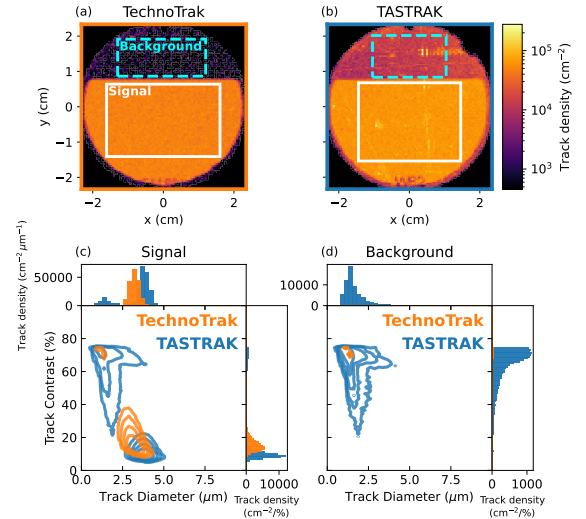


FIG. 2. Spatial histograms of the detected track density on (a) TechnoTrak and (b) TASTRAK CR-39 exposed to DD fusion protons ranged down to 2.6 MeV in the "signal" region (white) and completely blocked in the "background" region (teal). Track contrast and diameter contour plots and histograms of track density in the (c) "signal" region and (d) "background" region for TechnoTrak (orange) and TASTRAK (blue) CR-39.

20 cm away from the target and at 124° from the deuteron beam. The TASTRAK and TechnoTrak CR-39 were simultaneously exposed to either DD protons ranged down by flat Al filters to 0.9 or 2.6 MeV, or to D³He protons ranged down to 5.6 MeV. The above energies are given for protons produced at 90° to the deuteron beam. Protons produced at 107° have an ≈ 0.1 MeV downshift, as indicated by a kinematic beam-target physics simulation³⁰. The CR-39 was exposed until the proton track density reached $\sim 4 \times 10^4 \text{ cm}^{-2}$, as estimated by the SBD. The exposures for the 0.9, 2.6, and 5.6 MeV protons took 20, 10, and 90 min, respectively. Thick slabs of Al partially covered each piece of CR-39 to range out all charged fusion products which provide a background region. Figure 2(a,b) identifies the regions corresponding to signal and background on the scanned CR-39.

Following proton exposure, the CR-39 was placed in a bath of a 6N solution of NaOH at a temperature of 80°C to develop the tracks. The surface of the sample was then scanned with an optical microscope with a 40× objective lens. Automatic scanning software developed at MIT identifies tracks in microscope images by fitting an ellipse to round features in the image. The location, mean diameter, eccentricity, and optical contrast (100% is invisible, 0% is maximally visible) of each track is recorded. This scanning software is in use for all CR-39 diagnostics at the National Ignition Facility, the OMEGA Laser System, and the Z Pulsed Power Facility. The pieces exposed to 0.9, 2.6, and 5.6 MeV protons were etched for 2 hr, 3 hr, and 4 hr, respectively. Etch times were chosen in an effort to fully develop all tracks.

The spatial maps of track density of the two CR-39 variants exposed to 2.6 MeV protons are shown in Figure 2(a,b). The density of tracks in contrast-diameter space is shown in histograms for both the signal and background regions. Tracks are only accepted in our analysis if the eccentricity is less than 15% (0% is fully round); this cuts out significant noise and

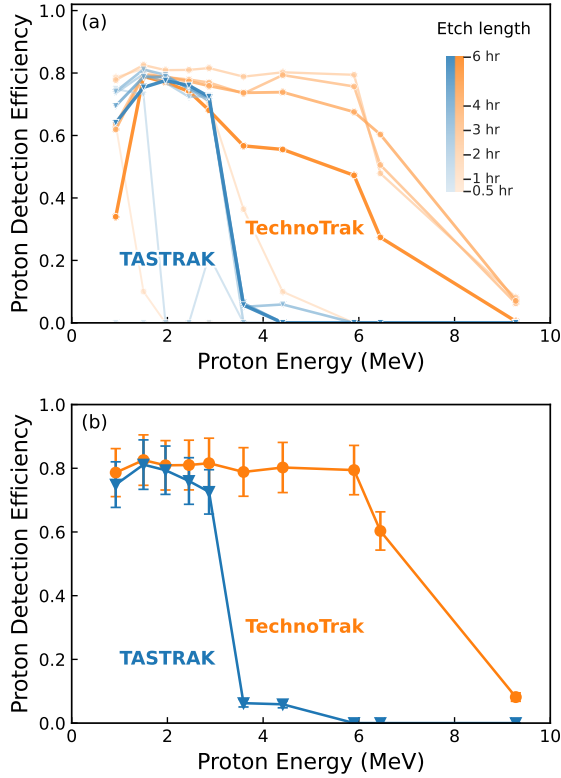


FIG. 3. (a) Measured proton detection efficiency of TechnoTrak (orange) and TASTRAK (blue) CR-39 as a function of incident proton energy shown for each etch length. (b) The maximum proton detection efficiency between all etch lengths for a given energy. Error bars include Poisson counting uncertainty and uncertainties in the SBD and CR-39 solid angles.

leaves most tracks untouched. In the background region, the scanning software falsely identifies unrelated aberrations in the samples as proton tracks. It is shown in Figure 2(d) that these false tracks are largely high contrast (nearly invisible) and low diameter. The TASTRAK CR-39 exhibits far more noise than the TechnoTrak CR-39, and across a wider gamut of contrast and diameter. The signal tends to exist in localized regions of contrast-diameter space as shown without any background subtraction in Figure 2(c).

The signal-to-background ratio (S/B) quantifies the level of intrinsic noise, and it can be computed within the island of true signal in contrast-diameter space. For proton energies of 0.9, 2.6, and 5.6 MeV, we find that the TechnoTrak CR-39 improves the signal-to-background ratio over TASTRAK CR-39 by factors of 4, 8, and 36, respectively. The detailed S/B calculations are discussed in Appendix A. The improvement in S/B is likely due to the TechnoTrak CR-39 having a particularly smooth surface because it is made from a purified monomer²⁹.

B. Efficiency & dynamic range

A second set of experiments on the linear ion accelerator measured the efficiency and dynamic range of both TechnoTrak and TASTRAK CR-39. These same experiments also measured the mean track diameter as a function of proton en-

ergy and etch time; these results are discussed in Section II C. A stepped Al filter with two rows of five filters each for a total of ten filters, each of different thickness, was placed in front of the CR-39 sample. A schematic of the filter pack is drawn in Figure 1(a) and a photograph of the setup for these experiments is shown in Figure 1(c). The five windows on the filter's top row range 3.0 MeV DD protons down to mean energies between 0.92 and 2.87 MeV, and on the bottom row range 14.7 MeV D³He protons down to mean energies between 3.59 and 9.28 MeV. The filter thicknesses and mean proton energies are written in Figure 1(b); this calibration is presented in previous work²³. The bottom row of filters completely ranges out the DD protons, and the top row ranges D³He protons down to a minimum mean energy of 14.2 MeV, an energy to which CR-39 is insensitive. All filter steps range out the generated alpha particles and tritons; CR-39 is also relatively insensitive to the fusion neutrons⁷. A thick aluminum shutter is used to independently expose the CR-39 samples behind the top five windows to DD protons and those behind the bottom five windows to D³He protons; this is needed because of the factor of 20 difference in emission rate as mentioned above.

In separate exposures, circular samples (5 cm diameter) of both TASTRAK and TechnoTrak CR-39 were placed 15 cm from the target at an angle 90° from the deuteron beam. A silicon surface barrier detector (SBD) was placed 18 cm from the target at an angle 141° from the deuteron beam behind a 10 μ m Ta filter and an aperture with 0.17 cm² area. The CR-39 samples were exposed until the proton density reached $\sim 1.4 \times 10^4$ cm⁻², as estimated by the SBD. Each sample of CR-39 was exposed to DD and D³He protons for approximately 5 min and 100 min, respectively. In sum, each sample was under vacuum for approximately three hours, not long enough for the vacuum exposure to affect sensitivity²⁵. After exposure to protons, the CR-39 samples were etched in stages for 0.5, 1, 2, 3, 4, and 6 hr in cumulative time and scanned at each step. Each stage followed the same etch and scan procedures described previously in Section II A.

The number of tracks in each window of CR-39 is counted by integrating over a manually selected island of signal in contrast-diameter space after background subtraction and after removing tracks with an eccentricity beyond 15%. The track density is recorded for each window at each stage of the etching process. Efficiency is defined as the recorded track density on the CR-39 divided by the expected track density on the CR-39 measured by the SBD (accounting for the differences in detector solid angle).

Figure 3(a) shows the proton detection efficiency of TechnoTrak and TASTRAK CR-39 for each proton energy and etch length. Figure 3(b) shows, for each proton energy, the maximum efficiency attained by the CR-39 variants across all etch lengths. The values plotted in Figure 3(b) represent the efficiency of the CR-39 assuming one etches the CR-39 to the optimal length for track development at a given proton energy. Error bars include Poisson counting uncertainty and uncertainties in the SBD and CR-39 solid angles. The highest recorded efficiency was $83 \pm 8\%$ achieved on TechnoTrak CR-39 with 1.5 MeV protons etched for 2 hours. CR-39 is typically expected to have 100% detection efficiency for protons ~ 2 MeV⁴. Discrepancy between this measurement and CR-39's expected 100% efficiency may be due to a systematic error in SBD proton counting, cutting out higher eccentricity tracks from off-normal protons, or overlapped tracks not be-

ing counted. Only 0.1 – 0.9% of tracks used in the efficiency measurement are expected to be overlapped based on mean track density and diameter. Systematic counting error in the SBD used in this experiment has been previously observed to differ by approximately 10%³¹.

TechnoTrak CR-39 attains near perfect efficiency at detecting protons between 0.92 and 5.9 MeV, and signal is still detected from protons as high as 9.28 MeV. Using an identical analysis procedure, TASTRAK CR-39 attains near perfect detection efficiency for protons up to 2.87 MeV, but efficiency steeply falls off beyond 2.87 MeV. Any potential signal on TASTRAK CR-39 was indistinguishable from noise beyond 5.9 MeV. It is believed that the antioxidant ingredient contained in TechnoTrak CR-39 is largely responsible for the high dynamic range TechnoTrak CR-39 exhibits. TechnoTrak's low noise floor likely also plays a role in the high maximum detectable energy. Curiously, TechnoTrak CR-39 exhibits a significant decrease in efficiency when etched for 6 hours; the cause of this decrease is currently unknown.

C. Track diameter as a function of proton energy and etch time

The mean diameter of proton tracks on TechnoTrak and TASTRAK is measured for ten proton energies between 0.92 and 9.28 MeV and six etch lengths between 0.5 and 6 hr using the previously described procedure.

The dependence of proton track diameter on incident proton energy has been extensively characterized and modeled in the literature^{4,23,24}. Track diameter and proton energy are known to have an inverse relationship, and the specific relationship can be used to infer a proton spectrum from track diameters^{17,18}. Track diameter is also known to monotonically increase with etch time because the damaged material in the track etches away faster than the bulk material.

Our data reproduce these results for TechnoTrak CR-39 and show that the mean diameter of proton tracks on TechnoTrak CR-39 is either the same or lower than on TASTRAK CR-39 for protons below 2.5 MeV. Figure 4 shows the mean diameter of tracks as a function of proton energy and etch length for each CR-39 variant. The track diameter at early etch time is initially the same for each piece, but at longer etch times the difference in track sizes between TechnoTrak and TASTRAK CR-39 becomes larger. For protons above 2.5 MeV, TechnoTrak CR-39 has the larger track diameter, although data are limited because the efficiency of TASTRAK CR-39 falls off. Track diameter d can be expressed in terms of track etch rate V_T , bulk etch rate V_B , and etch time t as³²

$$d(t) = 2V_{Bt} \sqrt{\frac{V_T/V_B - 1}{V_T/V_B + 1}}.$$

The data then indicate that TechnoTrak CR-39 has different bulk and track etch rates than TASTRAK CR-39.

III. CONCLUSIONS AND FUTURE WORK

Two sets of experiments were completed to study the performance of Chiyoda Technol Corporation's TechnoTrak CR-39 in detecting MeV protons. Comparison is made to the

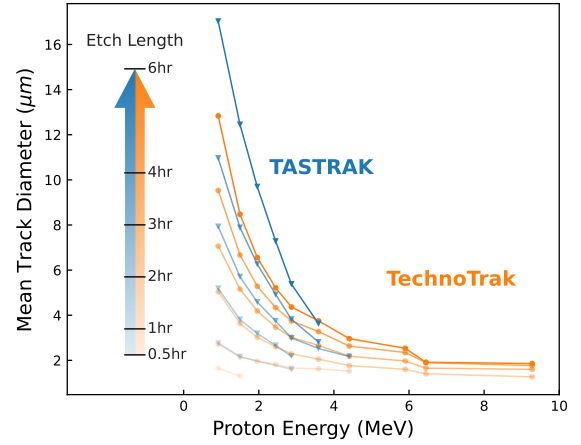


FIG. 4. The mean diameter of proton tracks as a function of incident proton energy on TASTRAK (blue) and TechnoTrak (orange) CR-39 for various etch times. One sample of each CR-39 variety with a ten-step Al filter was etched and scanned at etch times from 0.5 hr to 6 hr. The etch time is indicated by color lightness shown by the arrow pointing towards longer etch time. Diameter monotonically increases with etch time. Missing data points indicate windows in which proton tracks were indistinguishable from noise.

widely used TASTRAK CR-39 produced by Track Analysis Systems Ltd. CR-39 samples were exposed to DD and D³He fusion protons in the MIT linear ion accelerator. The first set of experiments studied the intrinsic noise; monoenergetic protons were exposed to large areas of each CR-39 variant simultaneously, and the identified tracks in the region exposed to protons were compared to that in a region with no nominal signal. In the second set of experiments, a ten-step Al filter and staged etching procedure were used to measure the proton detection efficiency for a broad range of proton energies and the diameter-to-energy relationship, for each variant of CR-39.

It is found that TechnoTrak CR-39 exhibits low intrinsic noise and a high dynamic range for proton detection. For proton energies of 0.9, 2.6, and 5.6 MeV, the signal-to-background ratio of TechnoTrak CR-39 exceeds that of TASTRAK CR-39 by factors of 4, 8, and 36, respectively. TechnoTrak CR-39 maintains near perfect detection efficiency for protons from 0.92 to 5.9 MeV, and still produces detectable tracks from 9.28 MeV protons. TASTRAK CR-39 maintained high proton detection efficiency only up to 2.87 MeV and signal became indistinguishable from noise above 5.9 MeV. The mean diameter of proton tracks on TechnoTrak CR-39 follows the expected inverse dependence on proton energy and monotonically increase with etch time. The proton track diameter on TechnoTrak CR-39 is smaller than that on TASTRAK CR-39 under comparable conditions for protons under 2.5 MeV. Above 2.5 MeV, TechnoTrak CR-39 has larger tracks than TASTRAK CR-39, although data are limited. The track diameter measurements indicate that TechnoTrak and TASTRAK CR-39 have differing bulk and track etch rates. Our study indicates that TechnoTrak CR-39 is a promising detector for current or future diagnostics used on many HED and ICF experiments requiring low background levels and high sensitivity over a broad proton energy range.

Future work should be done to further characterize the be-

havior and properties of TechnoTrak CR-39. The surface quality should be observed in detail to further explain the observed low intrinsic noise. The proton track diameter-to-energy relationship should be more precisely quantified using multiple samples of CR-39. The response to other charged particles, including alphas and deuterons, should also be studied. Additionally, the bulk and track etch rates of TechnoTrak CR-39 should be measured, as these values are important for models of the track diameter-to-energy relationship²⁴. Finally, HED and ICF diagnostics that currently use TASTRAK CR-39 should be tested and qualified with TechnoTrak CR-39 to verify that nominal results can be reproduced with TechnoTrak CR-39. The robustness of TechnoTrak CR-39 under realistic operating conditions should also be tested, including material aging during room-temperature versus sub-freezing storage, vacuum exposure, and x-ray exposure. Current standard operating procedures for TASTRAK CR-39 may differ from those for TechnoTrak CR-39, and should be adjusted as needed.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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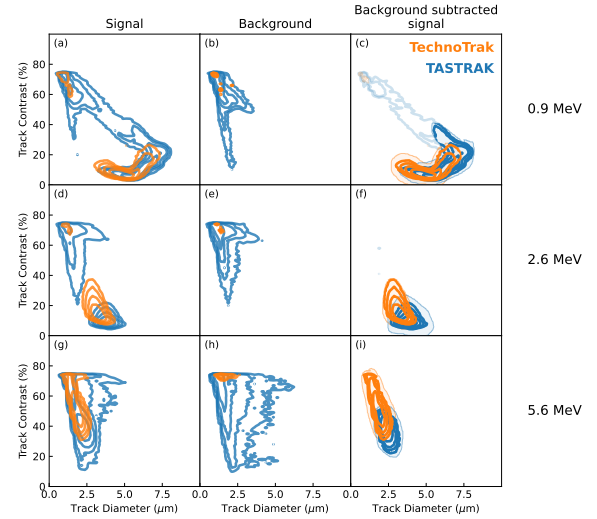


FIG. 5. The contours of track density in contrast-diameter space for (a,d,g) the signal region, (b,e,h) the background region, and (c,f,i) the signal region after a background subtraction. The contours are shown for each proton energy: (a,b,c) 0.9 MeV, (d,e,f) 2.6 MeV, and (g,h,i) 5.6 MeV for both TASTRAK (blue) and TechnoTrak (orange).

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Appendix A: Signal-to-background computation

The contrast-diameter contours of the signal and background regions for both TASTRAK and TechnoTrak CR-39 for each proton energy the samples were exposed to are shown in Figure 5. The background subtracted signal contours are also shown with manually identified islands of signal in contrast-diameter space.

The level of intrinsic noise is quantified by a signal-to-background (S/B) metric. The track density in the signal region of the CR-39 is subtracted by the track density in the background region in contrast-diameter space. This background subtracted data represents the true signal. Dividing

the integrated true signal (S) by the integrated background (B) yields an S/B value. We compute S/B twice: (1) separately integrating the true signal and the background over all contrast-diameter space, and (2) integrating over only the manually identified island of signal in contrast-diameter space. The computed values for each incident proton energy and CR-39 variant are written in Table I. Integrating the signal and background over the entire range of contrast-diameter space is often unhelpful because it is often easy to separate the island of signal from the majority of the background. Additionally, typical CR-39 analysis procedures integrate only over the island of signal. Hence, we compute S/B within the signal island.

Proton Energy	CR-39	S/B (all c, d)	S/B (signal island)
0.9 MeV	TechnoTrak	59	2000
0.9 MeV	TASTRAK	2.4	460
2.6 MeV	TechnoTrak	73	2600
2.6 MeV	TASTRAK	3.1	340
5.6 MeV	TechnoTrak	22	41
5.6 MeV	TASTRAK	0.24	1.2

TABLE I. Signal-to-background ratios for different proton energies over all contrast-diameter (c, d) space and over the selected signal island in contrast-diameter space.

The TechnoTrak CR-39 has ~ 25 times higher S/B than the TASTRAK CR-39 for 0.9 and 2.6 MeV protons, and ~ 90 times higher S/B for 5.6 MeV protons, when integrating over all contrast-diameter space. When integrating over only the island of signal, which is more practically relevant, the TechnoTrak CR-39 outperforms the TASTRAK CR-39 by 4, 8, and 34 times for 0.9, 2.6, and 5.6 MeV protons, respectively.